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NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Graph Algorithms (course)

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Course
outlineAbout
NPTEL ()How does an
NPTEL
online
course
work? ()Week 1
Introduction
and
Principles of
Algorithms ()Week 2
Shortest
Path
Algorithms
and
problems ()Week 3
Bellman
Equations ()

Week 5 Assignment 5

The due date for submitting this assignment has passed.

Due on 2025-08-27, 23:59 IST.

Assignment submitted on 2025-08-27, 16:00 IST

1) Consider the graph with $V=\{1,2,3\}; E=\{(1,2),(2,1),(1,3),(3,1),(2,3)\}$ with weights $\{2,8,9,1,6\}$ respectively. Solve APSP problem on V by using Floyd-Warshall algorithm and answer following: **1 point**

The first row of A_0 is {_,_,_}

- ☒ {0,2,9}
- ☐ {2,0,9}
- ☐ {0,9,2}
- ☐ {9,2,0}

Yes, the answer is correct.

Score: 1

Accepted Answers:

{0,2,9}

2) Consider the graph with $V=\{1,2,3\}; E=\{(1,2),(2,1),(1,3),(3,1),(2,3)\}$ with weights $\{2,8,9,1,6\}$ respectively. Solve APSP problem on V by using Floyd-Warshall algorithm and answer following: **1 point**

The second row of A_0 is {_,_,_}

- ☐ {6,0,8}
- ☒ {8,0,6}
- ☐ {8,6,0}
- ☐ {0,8,6}

Yes, the answer is correct.

Week 4
Bellman
Ford and
Dijkstra
Algorithm ()

Week 5 All
Pair Shortest
Path ()

☐ All Pair
Shortest Path
(unit?
unit=22&lesso
n=60)

☐ All Pair
Shortest Path
2 (unit?
unit=22&lesso
n=61)

☐ All Pair
Shortest Path
3&4 (unit?
unit=22&lesso
n=62)

☐ All Pair
Shortest Path
5 (unit?
unit=22&lesso
n=63)

☐ Week 5
Lecture
Material (unit?
unit=22&lesso
n=66)

☒ **Quiz: Week 5**
Assignment 5
(assessment?
name=101)

☐ Feedback
Form (unit?
unit=22&lesso
n=30)

Week 6
Prims and
Kruskal's
Algorithm ()

Week 7 DFS
and Cut

Score: 1

Accepted Answers:

{8,0,6}

3) Consider the graph with $V=\{1,2,3\}$; $E=\{(1,2),(2,1),(1,3),(3,1),(2,3)\}$ with weights {2,8,9,1,6} respectively. Solve APSP problem on V by using Floyd-Warshall algorithm and answer following:

1 point

The first row of A_1 is {_,_,_}

☐ {2,0,9}

☐ {0,9,2}

☐ {9,2,0}

☒ {0,2,9}

Yes, the answer is correct.

Score: 1

Accepted Answers:

{0,2,9}

4) Consider the graph with $V=\{1,2,3\}$; $E=\{(1,2),(2,1),(1,3),(3,1),(2,3)\}$ with weights {2,8,9,1,6} respectively. Solve APSP problem on V by using Floyd-Warshall algorithm and answer following: The third row of A_1 is {_,_,_}

1 point

☐ {1,0,3}

☐ {0,3,1}

☒ {1,3,0}

☐ {3,1,0}

Yes, the answer is correct.

Score: 1

Accepted Answers:

{1,3,0}

5) Consider the graph with $V=\{1,2,3\}$; $E=\{(1,2),(2,1),(1,3),(3,1),(2,3)\}$ with weights {2,8,9,1,6} respectively. Solve APSP problem on V by using Floyd-Warshall algorithm and answer following:

1 point

The first row of A_2 is {_,_,_}

☐ {2,0,8}

☒ {0,2,8}

☐ {8,2,0}

☐ {0,8,9}

Yes, the answer is correct.

Score: 1

Accepted Answers:

{0,2,8}

6) Consider the graph with $V=\{1,2,3\}$; $E=\{(1,2),(2,1),(1,3),(3,1),(2,3)\}$ with weights {2,8,9,1,6} respectively. Solve APSP problem on V by using Floyd-Warshall algorithm and answer following:

1 point

The second row of A_3 is {_,_,_}

☒ {7,0,6}

Vertex ()

Video
download ()Live session
()☐ {0,7,6}☐ {7,2,0}☐ {0,7,9}

Yes, the answer is correct.

Score: 1

Accepted Answers:

{7,0,6}

7) The complexity of Floyd-Warshall Algorithm for a graph G with b vertices and n edges is $\theta(_)$ **1 point**

☐ $n^{1/3}$ ☐ $n^{1/2}$ ☒ n^3 ☐ n^2

Yes, the answer is correct.

Score: 1

Accepted Answers:

 n^3

8) In converting the APSP problem to a matrix multiplication problem, we use special interpretations for matrix operation. **1 point**

The operation '+' is interpreted as ____

☒ Minimum☐ Maximum☐ Neither minimum nor maximum☐ Both minimum and maximum

Yes, the answer is correct.

Score: 1

Accepted Answers:

Minimum

9) In converting the APSP problem to a matrix multiplication problem, we use special interpretations for matrix operation. **1 point**

The operation '.' is interpreted as ____

☒ Addition☐ Multiplication☐ Dot product☐ None of the options given here

Yes, the answer is correct.

Score: 1

Accepted Answers:

Addition

10) The complexity of Johnson's Algorithm for APSP is $\theta(_)$ **1 point**



$$n \log n + nm$$



$$n^2 \log n + nm$$



$$n^2 \log n - nm$$



$$n^2 \log nm + nm$$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$$n^2 \log n + nm$$

